| DPP | DEPARTMENT POLICY AND PROCEDURE | | |
|---|---|---|---|
| Title | **CARE OF PEDIATRIC WITH PNEUMOTHORAX** | | |
| Department | **Pediatric Intensive Care Unit** | | |
| Applies to | **All PICU Medical and Nursing Staff** | | |
| Policy Code and Version Number | **DPP-PICU-004 (v2)** | | |
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| | NAME | POSITION | SIGNATURE | DATE |
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## 1.0 PURPOSE

1.1. To unify the nursing care to pediatric patients with Pneumothorax.

1.2. To be able to evacuate *j* drain accumulated air of fluid in the pleural space by insertion of chest tube.

1.3. To provide effective breathing pattern and promote effective gas exchange.

## 2.0 DEFINITION/ABBREVIATION

2.1. **Pneumothorax** is the presence of air in the pleural space occurring spontaneously or from trauma.

2.2. **Spontaneous pneumothorax** is a sudden onset of air in the pleural space with deflation of the affected lung in the absence of trauma.

2.3. **Open pneumothorax** an opening in the chest wall large enough to allow air to pass freely in and out of thoracic cavity with each attempted respiration.

2.4. **Tension pneumothorax** is a build of air under pressure in the pleural space resulting in interference with filling of both the heart and the lungs.

## 3.0 RESPONSIBILITY

3.1 **Physician** is responsible in assessing the patient condition and to order if chest tube will be inserted.

3.2 **Head Nurse** is responsible to make sure that needed equipments are always available in the unit and nurses are well knowledge about caring patients with pneumothorax.

3.3 **Charge Nurses** - responsible in double checking physician's orders and making sure that all equipments needed are ready; to help bedside nurse in case of chest tube insertion.

3.4 **Staff nurses** are responsible in frequent re-assessment of patient and inform physician if any changes on patient conditions and to prepare equipment needed if chest tube insertion is needed.

3.5 **Respiratory Therapist** is responsible to prepare intubation tray, mechanical ventilator and assist physician if intubation is indicated.

## 4.0 POLICY

4.1. The PICU staff must have knowledge about pneumothorax.

4.2. The PICU staff must be alert to assess for signs and symptoms of pneumothorax.

4.3. Equipment for needle aspiration and chest tube insertion should be ready in case needed.

4.4. Policy and procedure in the insertion and care of patient with chest tube should be followed.

4.5. Standard precaution shall be observed throughout the procedure.



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## 5.0 PROCEDURE (S)

### 5.1. *Preparatory phase*

5.1.1. Wash hands before attending the patient.

5.1.2. Prepare bed and other equipment for patient

5.1.3. Admit patient without delay and keep patient on pediatric railed bed.

5.1.4. Identify the patient by using two identifiers as per IPSG Goal 1

5.1.5. Attach patient to cardiac monitor and pulse oximeter. Monitor vital signs and record.

5.1.6. Place patient in radiant warmer and regulate the temperature control as needed.

### 5.2. *Performance phase*

5.2.1. Insert an Intravenous cannula and administer intravenous fluid solutions as ordered.

5.2.2. Monitor pulse oximetry and ABG.

5.2.3. Provide oxygen.

5.2.4. Assess for increase respiratory distress, cyanosis and desaturation, apnea and nasal flaring.

5.2.5. Assist the physician in endotracheal intubation and attach to mechanical ventilator as ordered.

5.2.6. Suction secretions as needed.

5.2.7. Auscultate chest for diminished breath sound and percuss for hyper resonance. Observe for chest asymmetry.

5.2.8. Note for abdominal distention.

5.2.9. Assist in chest x-ray procedure as ordered by the physician.

5.2.10. Assist the physician in needle aspiration to release air in the pleural space.

5.2.11. Prepare equipments for chest tube insertion and assist the physician in the procedure.

5.2.12. Implement care for patient with chest tube (refer to chest tube policy and procedure). Continuously monitor vital signs including blood pressure and oxygen saturation.

5.2.13. Perform cardiopulmonary resuscitation if needed.

5.2.14. After pneumothorax has been drained, evaluate patient's condition, vital signs, repeat chest x-ray as ordered.

5.2.15. Check ABG as ordered and regulate oxygen therapy accordingly.



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5.3. ***Follow-up phase***

5.3.1. Monitor patient to detect changes in the health condition

5.3.2. Document complete information on patient's condition and response to interventions rendered.

## 6.0 EQUIPMENTS

6.1. None

## 7.0 ATTACHMENT/FORMS

7.1. N/A

## 8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0024(V2) –Care of Patient with Endotracheal Tube– PMSH, 15/2/1439

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

## 9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer



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